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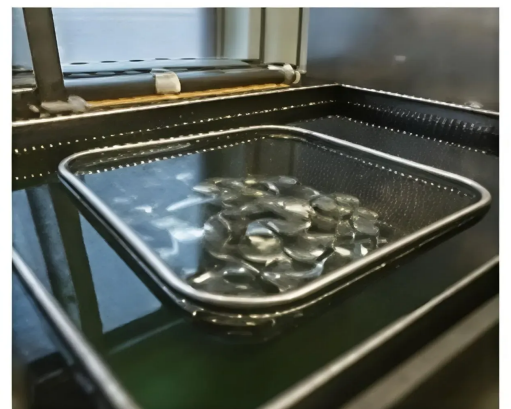
Collect information about "ELECTROPLATING". Make a list of articles in day-to-day life, where this technique is used.

Introduction:-

Electroplating is a process that uses an electric current to deposit a thin layer of metal onto the surface of another material. This technique is widely used in various industries to enhance the appearance, durability, and resistance to corrosion of different objects. The basic principle involves using an electrolyte solution containing metal ions and two electrodes: the object to be plated (cathode) and a metal anode. When electric current passes through the solution, metal ions are reduced and deposited on the cathode.

◆Key Steps in Electroplating :-

1. Surface Preparation :-



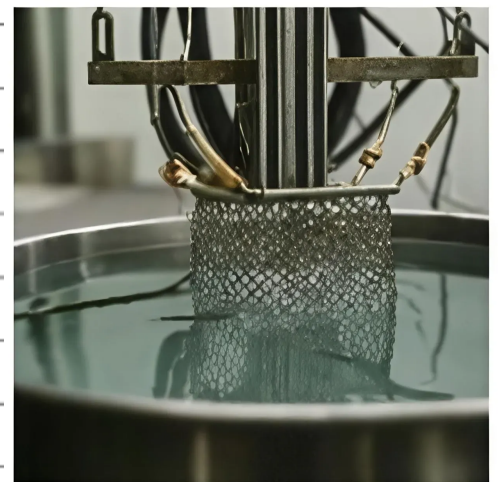
The object to be plated is cleaned thoroughly to remove dirt, grease, and other contaminants.

2. Electrolyte Solution :-



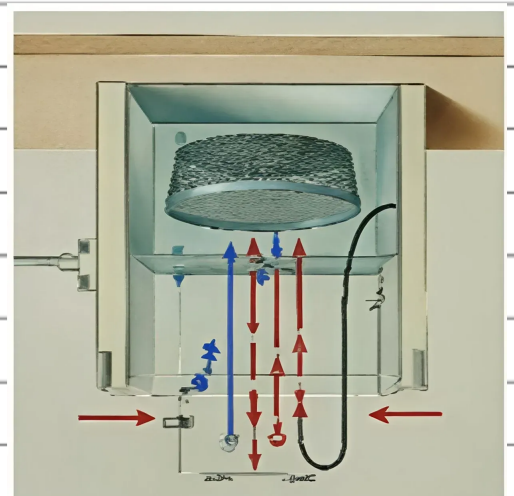
A solution containing metal salts (e.g., nickel sulfate for nickel plating) is prepared.

3. Electrode Setup :-



The object to be plated (cathode) and a metal piece (anode) are immersed in the electrolyte solution.

4. Electric Current :-



A direct current (DC) is passed through the solution, causing metal ions to migrate to and coat the cathode.

◆ Day-to-Day Articles Using Electroplating :-

Electroplating is employed in numerous everyday items to improve their properties and aesthetics. Here is a list of such articles:

1. Jewelry :-



- Gold-Plated Jewelry :- Enhances appearance and provides a luxurious finish.
- Silver-Plated Cutlery :- Improves aesthetic appeal and corrosion resistance.

2. Household Items :-



- Faucets and Fixtures :- Often chrome-plated for a shiny, durable finish.
- Door Handles and Knobs :- Nickel or brass-plated for durability and appearance.

3. Electronics :-



- Printed Circuit Boards (PCBs) :- Copper electroplating to create conductive pathways.
- Connectors and Contacts :- Gold or silver-plated for improved conductivity and corrosion resistance.

4. Automotive Parts :-



- Bumpers and Trim :- Chrome-plated for corrosion resistance and aesthetic appeal.
- Engine Parts :- Nickel or chromium-plated to reduce wear and enhance performance.

5. Coins and Medals :-



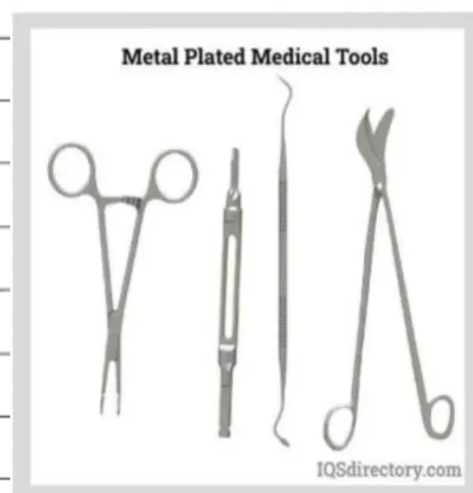
- Commemorative Coins :- Often plated with precious metals like gold or silver.
- Medals and Trophies :- Plated to enhance appearance and value.

6. Kitchenware :-



- Cutlery and Utensils :- Silver or nickel-plated for aesthetic and corrosion resistance.
- Cookware :- Copper or nickel-plated for even heat distribution and durability.

7. Medical Devices:-



- Surgical Instruments :- Nickel or chrome-plated to improve durability and sterilization.
- Orthopedic Implants :- Titanium or other biocompatible metal plating for improved integration with body tissues.

8. Hardware and Tools :-



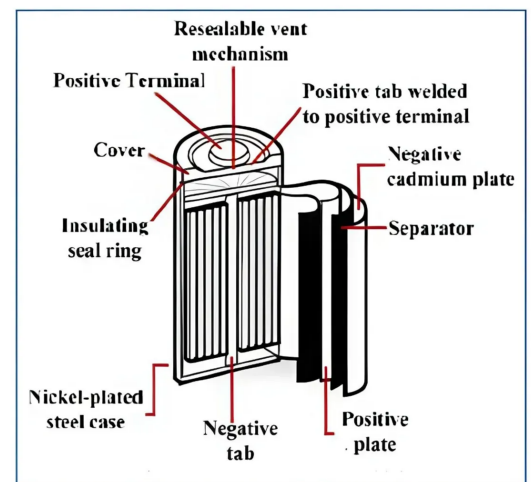
- Screws, Nails, and Bolts :- Zinc-plated to prevent rust and corrosion.
- Hand Tools :- Chrome or nickel-plated for increased hardness and corrosion resistance.

9. Watches and Timepieces:-



- Watch Cases and Bands:- Gold, silver, or rhodium-plated for appearance and durability.

10. Batteries:-



- Electrodes :- Nickel or cadmium-plated to enhance conductivity and efficiency.

❖ Advantages of Electroplating:-

1. Improves the appearance of objects by giving them a shiny and attractive finish.
2. Protects metals from rust and corrosion, increasing their life.
3. Increases the hardness and durability of metal surfaces.
4. Enhances electrical conductivity in electronic items.

❖ Disadvantages of Electroplating:-

1. Electroplating chemicals can cause environmental pollution.
2. The process can be expensive due to the use of costly metals.
3. Improper electroplating may peel or crack after some time.
4. Exposure to harmful chemicals can affect human health.

❖ References:-

1. Maharashtra State Board Science Textbook
2. NCERT Science Book
3. Encyclopaedia Britannica - Electroplating
4. Wikipedia - Electroplating

Conclusion:-

Electroplating is a versatile and essential technique used in many aspects of daily life to improve the appearance, durability, and performance of a wide range of products. From enhancing the aesthetic appeal of jewelry and household fixtures to improving the functionality of electronics and automotive parts, electroplating plays a crucial role in modern manufacturing and product design.